Finite Gain Graphs: A Verified Projective-Density Control, a Histories-Consistency Obstruction, and an Additive Holonomy Counterexample

Director A, fourth grounding supplement

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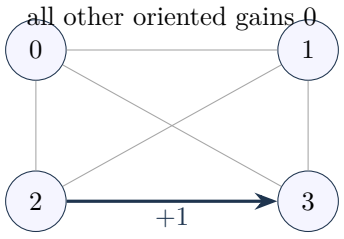
Abstract

We inspect directly the second-session papers G7 and G8 in the supplied Z2 edition and the B3/G3 packets. We give an explicit four-vertex *simple* gain graph whose cyclic covering graph is primitive, thereby satisfying the walk-count growth premise of G8 Proposition 5.1 for every cyclic gain order. With a separately specified faithful complex character, the event rays are dense in $\mathbb{CP}^3$; the graph still fails the natural projective consistency requirement for a single histories decoherence functional across walk lengths. An exact cross-event witness rules out repair by scalar normalization. We also construct a gauge-invariant, nonnegative *additive* holonomy-sensitive measure once terminal/root equality is available, precisely bounding G8's endpoint-blind no-go lemma. A sign slip in the *proof* of G7's determinant identity is repaired without changing the identity. These are conditional mathematics and a source erratum, not a derivation of physical probability, dynamics, classical mechanics, or an ontology.

Research status. Initial External Grounding, G1 OPEN. No total ontology selected. All finite trial rows are retained in the machine-readable companion; reported computations are not independent scientific certification.

1 Explicit carrier and primitive covering

For $n \geq 2$ take the complete graph K_4 on vertices $0, 1, 2, 3$, cyclic gain group $\mathbb{Z}_n$ (written additively), and gain $+1$ on $2 \to 3$, -1 on $3 \to 2$, and 0 on all other oriented edges. This is a *stipulated* abstract relation with no interpretation of arrows as physical time. The associated covering graph $\widetilde{\Gamma}_n$ has vertices (v, g) and edges $(u, g) \sim (v, g + \varphi(u, v))$. All its vertices have degree three.



Theorem 1.1 (A concrete growth witness)**.** *For every integer $n \geq 2$, $\widetilde{\Gamma}_n$ is connected and nonbipartite. Let $N_k(v,g)$ count k-edge walks beginning at $(0,0)$ and ending at (v,g). Then for each fixed v, g,*

$$N_k(v,g) = \frac{3^k}{4n} + O(\rho_n^k), \qquad 0 \leq \rho_n < 3. \tag{1}$$

In particular all $4n$ endpoint/gain counts tend to infinity simultaneously *along the entire integer sequence $k \to \infty$.*

Proof. The path $0 \to 2 \to 3 \to 0$ has net gain 1 and therefore its lifted repetitions reach every gain sector. Within a sector the lifts of paths whose net gain is 0 connect all four base vertices, so the covering graph is connected. The base triangle $0 \to 1 \to 2 \to 0$, whose gains all vanish, lifts to an odd cycle. Hence the cover is nonbipartite. Its adjacency matrix M_n is real symmetric, nonnegative, irreducible, aperiodic, and three-regular. Perron–Frobenius implies the eigenvalue 3 is simple and strictly dominates every other eigenvalue in modulus. The associated unit eigenvector is the constant vector divided by $\sqrt{4n}$. The spectral decomposition of M_n^k gives (1) entrywise. □

Corollary 1.2 (G8's dense-ray condition is realized explicitly)**.** *For $n \geq 3$, choose the faithful character $\chi(g) = e^{2\pi i g/n}$. For a walk event S, assign $v_S(y) = \sum_{w \in S,\, t(w)=y} \chi(\varphi(w))$. The union over all walk lengths k of the nonzero rays $[v_S]$ is dense in $\mathbb{CP}^3$ with its ordinary projective topology and transition-probability metric. This establishes an actual class of finite graph examples for the growth premise,* not *the physical meaning of the metric.*

Proof. By Theorem 1.1, for any finite prescribed nonnegative counts of walks with each endpoint and gain, there is one sufficiently large k where the counts can all be selected as subsets of actual walks. Since $1 + \chi(1) + \cdots + \chi(n-1) = 0$, every negative contribution to a cyclotomic integer can be replaced by a finite sum of positive root-of-unity contributions. Thus every vector in $\mathbb{Z}[\chi(1)]^4$ occurs at some k. The ring contains 1 and one nonreal number $\chi(1)$; integer combinations form a rank-two lattice when $n = 3$ (and in any case its field of fractions $\mathbb{Q}(\chi(1))$ contains a dense rational plane in $\mathbb{C}$). Rays through such vectors are dense in $\mathbb{CP}^3$. These event vectors are a *constructed* geometry, not identified preparations. □

Remark 1.3 (Exact quantitative control)**.** *For $n = 3$, the 12×12 cover matrix first has all entries strictly positive at its fourth power. Starting from $(0,0)$, the minimum of the 12 gain/endpoint counts at $k = 4, 8, 12$ is respectively $2, 348, 36126$; the totals are 3^k. Every row for $k = 0, \ldots, 18$ is included without filtering.*

2 An obstruction between successive walk lengths

For $n = 3$, put $\omega = e^{2\pi i/3}$. The represented adjacency operator, with columns indexed by sources, is the Hermitian matrix

$$A = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & \bar{\omega} \\ 1 & 1 & \omega & 0 \end{pmatrix}. \tag{2}$$

For an event of walks S of length k let $v_k(S)$ be the sum of its weighted endpoint vectors and $D_k(S,T) = \langle v_k(T), v_k(S) \rangle$. Let $\operatorname{Ext} S$ contain *all* one-edge extensions of paths in S. Then $v_{k+1}(\operatorname{Ext} S) = A v_k(S)$.

Proposition 2.1 (Failure of cylinder consistency)**.** *The kernels D_k cannot be made projectively consistent under* Ext *by multiplying each D_k by an arbitrary nonzero scalar depending only on k. In particular, they do not define a normalized decoherence functional on cylinder events of infinite walks by that identification.*

Proof. Let $S = \{0 \to 1\}$ and $T = \{0 \to 2\}$ at length one. Their endpoint vectors are e_1, e_2, hence $D_1(S,T) = 0$. The vectors of their complete extensions are $Ae_1 = (1,0,1,1)^T$ and $Ae_2 = (1,1,0,\omega)^T$, giving

$$D_2(\mathrm{Ext}\, S, \mathrm{Ext}\, T) = \langle Ae_1, Ae_2\rangle = 1 + \omega = \tfrac{1+i\sqrt{3}}{2} \neq 0. \tag{3}$$

No nonzero scalar normalization can turn the nonzero cross term into zero. Furthermore, for Ω_k the event of all k-walks, $\mu_0(\Omega_0) = 1, \mu_1(\Omega_1) = 3, \mu_2(\Omega_2) = 15$, since $A^2 e_0 = (3, 2, 1+\bar{\omega}, 1+\omega)^T$. Normalizing at each length by its total mass gives $\mu_1(S)/3 = 1/3$, but $\mu_2(\mathrm{Ext}\, S)/15 = 3/15 = 1/5$. Both diagonal and off-diagonal forms of consistency fail. □

Remark 2.2 (Scope)**.** *Projective cylinder consistency is an* additional requirement *when interpreting successive k as refinements of one history. G8 proves fixed-length strong positivity and a Hilbert-space quotient; it does not assert this cross-length requirement. Proposition 2.1 therefore identifies a missing dynamics/history bridge and does* not *refute G8's fixed-k results. One could add a unitary update, dilation, clock, or different event algebra, but each requires a separate construction and a warranted operational interpretation. The same K_4 graph supports the ordinary classical Markov cylinder probabilities $|S|/3^k$, illustrating nonselection: those classical probabilities* are *consistent under* Ext *but ignore the gains.*

3 Endpoint-aware first-order holonomy observables

G8 Lemma 3.1 correctly states that if one universal function $F_1 : G \to \mathbb{R}$ is applied to the *gain alone* for every walk on every graph, switching invariance makes F_1 constant. Dropping the endpoint-blindness hypothesis changes the assertion.

Proposition 3.1 (Positive, additive, gain-sensitive measure)**.** *For $G = \mathbb{Z}_3$ and root vertex x, define for every fixed-length event S*

$$\nu(S) = \sum_{w \in S} \mathbf{1}_{t(w)=x}(1 + \Re\chi(\varphi(w))). \tag{4}$$

Then ν is nonnegative, finite-additive, invariant under every vertex switching, and dependent on nontrivial closed-walk holonomy. Its diagonal intersection kernel $D^{\mathrm{cl}}(S,T) = \sum_{w \in S\cap T} \mathbf{1}_{t(w)=x}(1 + \Re\chi(\varphi(w)))$ is strongly positive. There is no second- or higher-order interference.

Proof. For $\mathbb{Z}_3$ the weight is 2 when $\varphi(w) = 0$ and $1/2$ when it is 1 or 2. Additivity follows from disjoint summation and strong positivity from the nonnegative diagonal Gram representation. Switching changes the gain of a walk from x to x by conjugation, which is trivial in $\mathbb{Z}_3$. In K_4, the closed three-step walks $0 \to 1 \to 2 \to 0$ and $0 \to 2 \to 3 \to 0$ have respective gains 0 and 1, hence weights 2 and $1/2$. An additive measure has $I_r = 0$ for every $r \geq 2$. □

This does not contradict G8 Lemma 3.1, since $\mathbf{1}_{t(w)=x}$ is not a function of gain alone. It *does* defeat an interpretation that switching invariance plus positivity, without endpoint-blindness, forces interference of order two before the transport becomes observable.

4 Proof erratum for G7, without changing its determinant law

Let $B = \prod_{j=0}^{k-1} \langle a_j \mid a_{j+1}\rangle$ with $a_k = a_0$ and $e^{i\theta} = B/|B|$. In the proof of G7 Proposition 4.3, consecutive horizontal phases obey $\alpha_{j+1} = \alpha_j - \arg\langle a_j \mid a_{j+1}\rangle$. Hence the *terminal* lift is

$$\widetilde{a}_k = e^{-i\theta} a_0, \qquad U a_0 = e^{-i\theta} a_0, \qquad T = e^{+i\theta} U|_{a_0^\perp}. \tag{5}$$

The source proof instead writes a positive terminal exponent and negative tangent conversion. Those two signs must be reversed. Since $\det U = 1$, we obtain

$$\det_{\mathbb{C}} T = e^{in\theta} \det(U|_{a_0^\perp}) = e^{i(n+1)\theta} = (B/|B|)^{n+1}. \tag{6}$$

The announced determinant law remains correct. For a unit qubit triangle with $B = (9+3i)/16$, the horizontal lift ends at phase $(3-i)/\sqrt{10}$ and the tangent holonomy has eigenvalue $(4+3i)/5$, in accord with this correction.

5 Reconciliation and ontology obligations

Result	Status	Boundary / essential import
K_4 cover counts	Proved, conditional data	Finite graph, assigned gains, graph covering, Perron–Frobenius
Dense $\mathbb{CP}^3$ rays	Proved in mathematical model	Faithful complex character, sum-of-walk-amplitudes map, ray equivalence
Cross-length D_k	Obstruction proved	The *natural* all-path extension; alternative history definitions not excluded
First-order ν	Proved countercontrol	Closed-walk endpoint information; falls outside G8 endpoint-blind lemma
G7 determinant	Theorem retained; proof patch	Source's horizontal-lift signs amended; no physical holonomy extracted
Operational measurement	Unresolved	Empirical preparation/readout, effects, frequency, calibration
Dynamics and time	Unresolved	Strongly consistent process, choice of update, clock, reversible evolution
Classical mechanics	Unresolved	Symplectic phase space, Hamiltonian flows and controlled limit

The results are mathematical reductions, not a proof that any stipulated prephysical primitive exists in nature. The graph measure and a classical counting measure remain alternative completions of the same finite data. Director G3 independently supplies further nonselection controls for mixed-state tomography; Director B3 supplies conditional local-instrument selection boundaries. Neither closes these emergence obligations.

Methods, provenance and source context

This supplement was generated from inspection of the *actual* G7/G8 PDFs in the user-supplied Z2 archive and the B3/G3 submissions, rather than the disposition alone. Exact SymPy identities, finite-cover enumeration for $n = 2, \ldots, 8$, and every $k = 0, \ldots, 18$ count for $n = 3$ appear in `verify_A4_independent.py` and its JSON. Direct B3 and G3 mathematical verifiers reran. G8's original program reran with its predicates passing; several floating-point entries differ in the last digits, so no byte-identical reproduction is asserted for those entries. Source baseline editions are left unchanged. The reviews and proofs remain AI-assisted and unrefereed, with no independent sub-agent or experimental measurement.

Primary literature used as scope checks, not as endorsement. F. Dowker, S. Johnston and R. Sorkin, *Hilbert Spaces from Path Integrals*, J. Phys. A 43 (2010) 275302, doi:10.1088/1751-8113/43/27/275302; F. Dowker, S. Johnston and S. Surya, *On Extending the Quantum Measure*, J. Phys. A 43 (2010) 505305, doi:10.1088/1751-8113/43/50/505305; T. Gelander, *Limits of Finite Homogeneous Metric Spaces*, Enseign. Math. 59 (2013), 195–206, doi:10.4171/LEM/59-1-8; H. Barnum, M. P. Müller and C. Ududec, *Higher-order interference and single-system postulates characterizing quantum theory*, New J. Phys. 16 (2014) 123029, doi:10.1088/1367-2630/16/12/123029.